TPS631000 1.5A Output Current, High-Power Density Buck-Boost Converter

1 Features

- 1.6V to 5.5V input voltage range
  - Device input voltage > 1.65V for start-up
- 1.2V to 5.3V output voltage range (adjustable)
- High output current capability, 3A peak switch current
  - 2A output current for V_{IN} ≥ 3V, V_{OUT} = 3.3V
  - 1.5A output current for V_{IN} ≥ 2.7V, V_{OUT} = 3.3V
- High efficiency over the entire load range
  - 8µA typical quiescent current
  - Power save mode
- 150mV load-step response at a 1A current step
- Peak current buck-boost mode architecture
  - Seamless mode transition with < 20mV output ripple
  - Forward and reverse current operation
  - Start-up into prebiased outputs
  - Fixed-frequency operation with 2MHz switching
- Safety and robust operation features
  - Overcurrent protection and short-circuit protection
  - Integrated soft start with active ramp adoption
  - Overtemperature protection and overvoltage protection
  - True shutdown function with load disconnect
  - Forward and backward current limit
- Small design size
  - Small 1µH inductor
  - Works with one 0805 output capacitor over the full V_{OUT} range
- Create a custom design using the TPS631000 with the WEBENCH® Power Designer

2 Applications

- System pre-regulator (smartphone, tablet, terminal, telematics)
- Point-of-load regulation (wired sensor, port/cable adapter, and dongle)
- Fingerprint, camera sensors (electronic smart lock, IP network camera)
- RF amplifier supply (smart sensors)
- Voltage stabilizer (datacom, optical modules, cooling and heating)

3 Description

The TPS631000 is a constant frequency peak current mode control buck-boost converter. The device has a 3A peak current limit (typical) and 1.6V to 5.5V input voltage range. The TPS631000 provides a power supply design for system pre-regulators and voltage stabilizers.

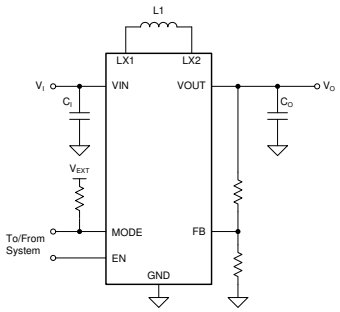
Depending on the input voltage, the TPS631000 automatically operates in boost, buck, or in 3-cycle buck-boost mode when the input voltage is approximately equal to the output voltage. The transitions between modes happen at a defined duty cycle and avoid unwanted toggling within the modes to reduce output voltage ripple. 8µA quiescent current and power save mode power enable the highest efficiency for light to no-load conditions.

The TPS631000 offers a very small design size with a 1.2mm × 2.1mm SOT-583 package, a 1µH inductor, and one 0805 output capacitor.

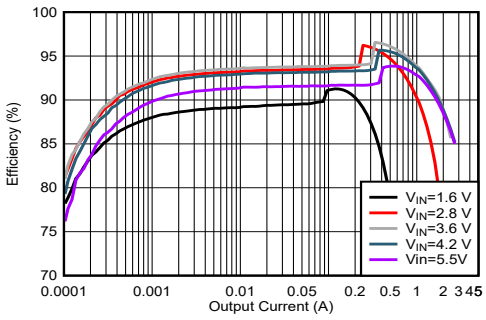
Package Information

PART NUMBER	PACKAGE(1)	PACKAGE SIZE(2)
TPS631000	DRL (SOT-5X3, 8)	2.1mm × 1.6mm

(1) For more information, see Section 10.
(2) The package size (length × width) is a nominal value and includes pins, where applicable.



Typical Application



Efficiency Target vs Output Current (V_{OUT} = 3.3V)

An IMPORTANT NOTICE at the end of this data sheet addresses availability, warranty, changes, use in safety-critical applications, intellectual property matters and other important disclaimers. PRODUCTION DATA.

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4 Pin Configuration and Functions

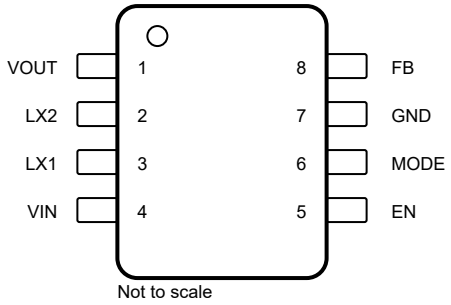


Figure 4-1. 8-Pin DRL SOT-5X3 Package (Top View)

Table 4-1. Pin Functions

PIN		TYPE[1]	DESCRIPTION
NAME	NO.		
VOUT	1	PWR	Power stage output
LX2	2	PWR	Inductor switching node of the boost stage
LX1	3	PWR	Inductor switching node of the buck stage
VIN	4	PWR	Supply input voltage
EN	5	I	Device enable. Set High to enable and Low to disable. This pin must not be left floating.
MODE	6	I	PFM/PWM selection. Set Low for power save mode, set High for forced PWM. This pin must not be left floating.
GND	7	PWR	Power ground
FB	8	I	Voltage feedback. Sensing pin

(1) PWR = power, I = input

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5 Specifications

5.1 Absolute Maximum Ratings

over operating junction temperature range (unless otherwise noted)(1)

		MIN	MAX	UNIT
V_I	Input voltage (VIN, LX1, LX2, VOUT, EN, FB, MODE)(2)	–0.3	6	V
	Input voltage for less than 10 ns (LX1, LX2)(2)	–2.0	7	V
T_J	Operating junction temperature	–40	150	°C
T_{stg}	Storage temperature	–65	150	°C

(1) Operation outside the Absolute Maximum Ratings may cause permanent device damage. Absolute Maximum Ratings do not imply functional operation of the device at these or any other conditions beyond those listed under Recommended Operating Conditions. If used outside the Recommended Operating Conditions but within the Absolute Maximum Ratings, the device may not be fully functional, and this may affect device reliability, functionality, performance, and shorten the device lifetime.
(2) All voltage values are with respect to network ground terminal, unless otherwise noted.

5.2 ESD Rating

			VALUE	UNIT
$V_{(ESD)}$	Electrostatic discharge	Human-body model (HBM), per ANSI/ESDA/JEDEC JS-001, all pins(1)	±1000	V
		Charged-device model (CDM), per ANSI/ESDA/JEDEC JS-002, all pins (2)	± 500	

(1) JEDEC document JEP155 states that 500-V HBM allows safemanufacturing with a standard ESD control process.
(2) JEDEC document JEP157 states that 250-V CDM allows safemanufacturing with a standard ESD control process.

5.3 Recommended Operating Conditions

over operating junction temperature (unless otherwise noted)

			MIN	NOM	MAX	UNIT
V_I	Supply voltage		1.6		5.5	V
V_O	Output voltage		1.2		5.3	V
C_I	Input capacitance	V_I = 1.6 V to 5.5 V	4.2			µF
C_O	Output capacitance	1.2 V ≤ V_O ≤ 3.6 V, nominal value at V_O = 3.3 V	10.4	16.9	330	µF
		3.6 V < V_O ≤ 5.3 V, nominal value at V_O = 5 V	7.95	10.6	330	µF
L	Inductance		0.7	1	1.3	µH
T_J	Operating junction temperature range		–40		125	°C

5.4 Thermal Information

over operating free-air temperature range (unless otherwise noted)

THERMAL METRIC(1)		TPS631000 DRL 8 PINS	UNIT
$R_{\Theta JA}$	Junction-to-ambient thermal resistance	132.7	°C/W
$R_{\Theta JC(top)}$	Junction-to-case (top) thermal resistance	43.8	°C/W
$R_{\Theta JB}$	Junction-to-board thermal resistance	27.3	°C/W
Ψ_{JT}	Junction-to-top characterization parameter	1.2	°C/W
Ψ_{JB}	Junction-to-board characterization parameter	26.6	°C/W
$R_{\Theta JC(bot)}$	Junction-to-case (bottom) thermal resistance	N/A	°C/W

(1) For more information about traditional and new thermal metrics, see the Semiconductor and IC Package Thermal Metrics application report.

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5.5 Electrical Characteristics

Over operating junction temperature range and recommended supply voltage range (unless otherwise noted). Typical values are at V_I = 3.8 V , V_O = 3.3 V and T_J = 25°C (unless otherwise noted).

PARAMETER			TEST CONDITIONS		MIN	TYP	MAX	UNIT
SUPPLY								
I_{SD}	Shutdown current into VIN		V_I = 3.8 V, $V_{(EN)}$ = 0 V	T_J = 25°C		0.5	0.9	µA
I_Q	Quiescent current into VIN		V_I = 2.2 V, V_O = 3.3 V, $V_{(EN)}$ = 2.2 V, no switching			0.15	6.1	µA
	Quiescent current into VOUT		V_I = 2.2 V, V_O = 3.3 V, $V_{(EN)}$ = 2.2 V, no switching			8		µA
V_{IT+}	Positive-going UVLO threshold voltage				1.5	1.55	1.599	V
V_{IT-}	Negative-going UVLO threshold voltage		During start-up		1.4	1.45	1.499	V
V_{hys}	UVLO threshold voltage hysteresis				99			mV
$V_{I(POR)T+}$	Positive-going POR threshold voltage		maximum of V_I or V_O		1.25	1.45	1.65	V
$V_{I(POR)T-}$	Negative-going POR threshold voltage				1.22	1.43	1.6	V
I/O SIGNALS								
V_{T+}	Positive-going threshold voltage	EN, MODE			0.77	0.98	1.2	V
V_{T-}	Negative-going threshold voltage	EN, MODE			0.5	0.66	0.76	V
V_{hys}	Hysteresis voltage	EN, MODE				300		mV
I_{IH}	High-level input current	(EN, MODE)	$V_{(EN)}$ = $V_{(MODE)}$ = 1.5 V, no pullup resistor			±0.01	±0.25	µA
I_{IL}	Low-level input current	(EN, MODE)	$V_{(EN)}$ = $V_{(MODE)}$ = 0 V,			±0.01	±0.1	µA
I_{Bais}	Input bias current	(EN, MODE)	$V_{(EN)}$ = 5.5 V			±0.01	±0.3	µA
POWER SWITCH								
$r_{DS(on)}$	On-state resistance	Q1	V_I = 3.8 V, V_O = 3.3 V, test current = 0.2 A			45		mΩ
		Q2				50		mΩ
		Q3				50		mΩ
		Q4				85		mΩ
$I_{Reverse}$	Reverse current into VOUT		V_I = 0 V, V_O = 3.3 V, $V_{(EN)}$ = 0 V	T_J = 25°C		0.8	2	µA
				T_J = –40°C to 125°C			12	µA
CURRENT LIMIT								
$I_{L(PEAK)}$	Switch peak current limit (1)	Q1	V_O = 3.3 V	Output sourcing current	2.6	3	3.35	A
				Output sinking current, V_I = 3.3 V	–0.7	–0.55	–0.45	A
I_{PFM_entry}	The output current at PFM mode entry threshold (peak) current (1)		I_O falling			145		mA
OUTPUT								
CONTROL[FEEDBACK PIN]								
V_{FB}	Reference voltage on feedback pin				495	500	505	mV
PROTECTION FEATURES								
$V_{T+(OVP)}$	Positive-going OVP threshold voltage				5.55	5.75	5.95	V
$V_{T+(IVP)}$	Positive-going IVP threshold voltage				5.55	5.75	5.95	V
TIMING PARAMETERS								
$t_{d(EN)}$	Delay between a rising edge on the EN pin and the start of the output voltage ramp					0.87	1.5	ms
$t_{d(ramp)}$	Soft-start ramp time				6.42	7.55	8.68	ms
f_{SW}	Switching frequency				1.8	2	2.2	MHz

(1) Current limit production test are performed under DC conditions. The current limit in operation is somewhat higher and depending on propagation delay and the applied external components

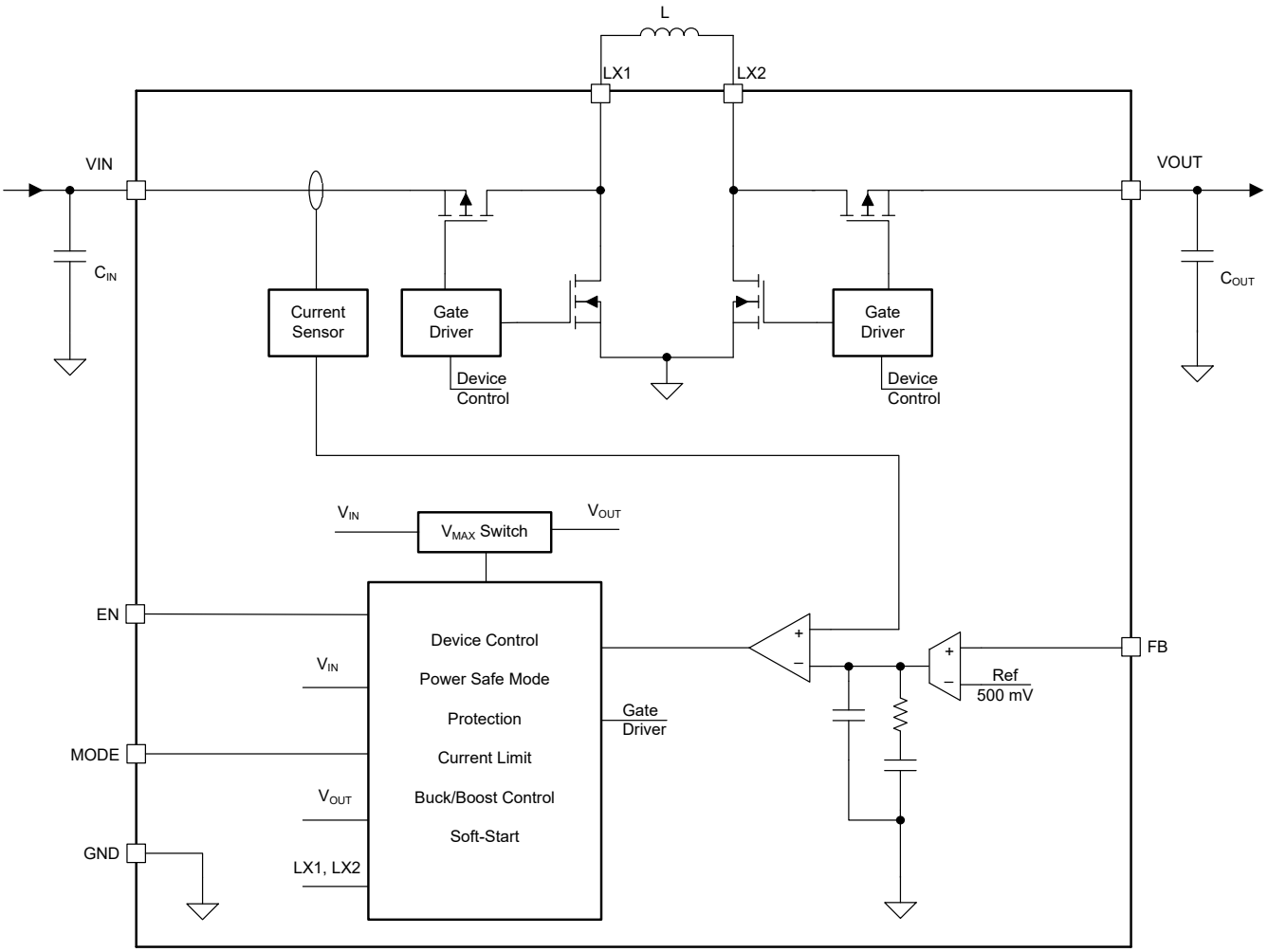
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6 Detailed Description

6.1 Overview

The TPS631000 is a constant frequency peak current mode control buck-boost converter. The converter uses a fixed-frequency topology with approximately 2MHz switching frequency. The modulation scheme has three clearly defined operation modes where the converter enters with defined thresholds over the full operation range of V_{IN} and V_{OUT}. The maximum output current is determined by the Q1 peak current limit, which is typically 3 A.

6.2 Functional Block Diagram



6.3 Feature Description

6.3.1 Undervoltage Lockout (UVLO)

The input voltage of the VIN pin is continuously monitored if the device is not in shutdown mode. UVLO only stops or starts the converter operation. The UVLO does not impact the core logic of the device. UVLO avoids a brownout of the device during device operation. In case the supply voltage on the VIN pin is lower that the negative-going threshold of UVLO, the converter stops the operation. To avoid a false disturbance of the power conversion, the UVLO falling threshold logic signal is digitally deglitched.

If the supply voltage on the VIN pin recovers to be higher than the UVLO rising threshold, the converter returns to operation. In this case, the soft-start procedure restarts faster than under start-up without a prebiased output.

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6.3.2 Enable and Soft Start

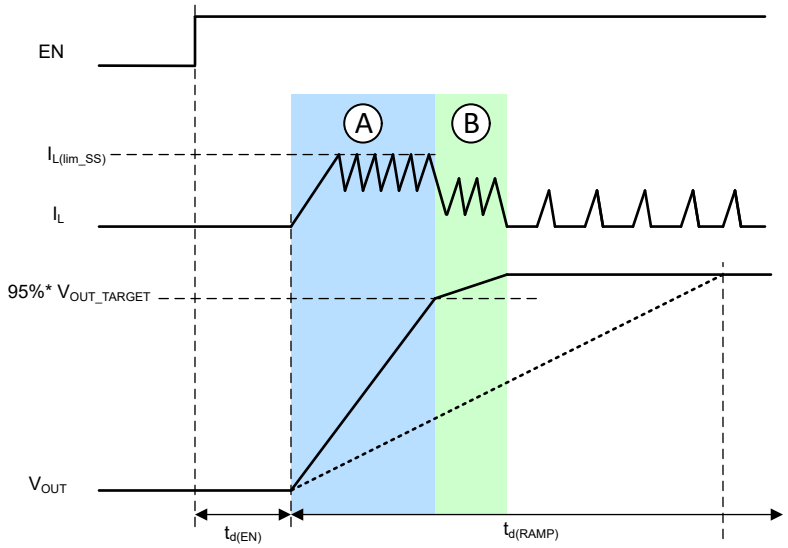


Figure 6-1. Typical Soft-Start Behavior

When the input voltage is above the UVLO rising threshold and the EN pin is pulled to a voltage above 1.2V, the TPS631000 is enabled and starts up after a short delay time, $t_{d(EN)}$.

The TPS631000 has an inductor peak current clamp to limit inrush current during start-up. When the minimum current clamp ($I_{L(lim_SS)}$) is lower than the current that is necessary to follow the voltage ramp, the current automatically increases to follow the voltage ramp. The minimum current limit ensures as fast as possible soft start if the capacitance is chosen lower than what the ramp time $t_{d(RAMP)}$ is selected for.

In a typical start-up case as shown in Figure 6-1 (low output load, typical output capacitance), the minimum current clamp limits the inrush current and charges the output capacitor. The output voltage then rises faster than the reference voltage ramp (see phase A in Figure 6-1). To avoid an output overshoot, the current clamp is deactivated when the output is close to the target voltage and follows the reference voltage ramp slew value given by the voltage ramp, which is finishing the start up (see phase B in Figure 6-1). The transition from the minimum current clamp operation is sensed by using the threshold 95% × V_{OUT_TARGET}. After phase B, the output voltage is well regulated to the nominal target voltage. The current waveform depends on the output load and operation mode.

6.3.3 Adjustable Output Voltage

The output voltage is set by an external resistor divider. The resistor divider must be connected between VOUT, FB, and GND. The feedback voltage is given by V_{FB}. The low-side resistor R2 (between FB and GND) must not exceed 100kΩ. Equation 1 calculates the high-side resistor R1 (between FB and VOUT).

$$R1 = R2 \times (V_{OUT} / V_{FB} - 1) \quad (1)$$

The typical V_{FB} voltage is 0.5V.

6.3.4 Mode Selection (PFM/FPWM)

The mode pin is a digital input to enable PFM/FPWM.

When the MODE pin is connected to logic low, the device works in auto PFM mode. The device features a power save mode to maintain the highest efficiency over the full operating output current range. PFM automatically changes the converter operation from CCM to pulse frequency modulation.

When the MODE pin is connected to logic high, the device works in forced PWM mode, regardless of the output current, to achieve minimum output ripple.

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6.3.5 Reverse Current Operation

The device can support reverse current operation (the current flows from VOUT pin to VIN pin). If the output feedback voltage on the FB pin is higher than the reference voltage, the converter regulation forces a current into the input capacitor. The reverse current operation is independent of the V_{IN} voltage or V_{OUT} voltage ratio, therefore it is possible on all device operation modes boost, buck, or buck-boost.

6.3.6 Protection Features

The following sections describe the protection features of the device.

6.3.6.1 Input Overvoltage Protection

The TPS631000 has input overvoltage protection. Input overvoltage protection avoids any damage to the device in case the current flows from the output to the input and the input source cannot sink current (for example, a diode in the supply path).

If forced PWM mode is active, the current can go negative until the current reaches the sink current limit. After the input voltage threshold, $V_{T+(IVP)}$, is reached on the VIN pin, the protection disables forced PWM mode and only allows current to flow from VIN to VOUT. After the input voltage drops under the input voltage protection threshold, forced PWM mode can be activated again.

6.3.6.2 Output Overvoltage Protection

The TPS631000 has the output overvoltage protection. Output overvoltage protection avoids any damage to the device in case the external feedback pin is not working properly.

If the output voltage threshold $V_{T+(OVP)}$ is reach on the VOUT pin, the protection disables converter power stage and enter a high impedance at the switch nodes.

6.3.6.3 Short Circuit Protection

The device features peak current limit performance at short circuit protection. Figure 6-2 shows a typical device behavior of an short/overload event of the short circuit protection.

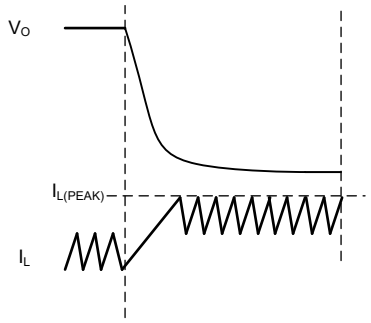


Figure 6-2. Typical Device Behavior During Short Circuit Protection

6.3.6.4 Thermal Shutdown

To avoid thermal damage of the device, the temperature of the die is monitored. The device stops operation once the sensed temperature rises over the thermal threshold. After the temperature drops below the thermal shutdown hysteresis, the converter returns to normal operation.

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6.4 Device Functional Modes

The device has two functional modes: off and on. The device enters the on mode when the voltage on the VIN pin is higher than the UVLO threshold and a high logic level is applied to the EN pin. The device enters the off mode when the voltage on the VIN pin is lower than the UVLO threshold or a low logic level is applied to the EN pin.

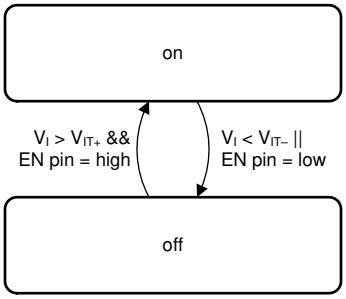


Figure 6-3. Device Functional Modes

7 Application and Implementation

Note

Information in the following applications sections is not part of the TI component specification, and TI does not warrant its accuracy or completeness. TI's customers are responsible for determining suitability of components for their purposes, as well as validating and testing their design implementation to confirm system functionality.

7.1 Application Information

The TPS631000 is a high-efficiency, low-quiescent current, buck-boost converter. The device is designed for applications needing a regulated output voltage from an input supply that can be higher or lower than the output voltage.

7.2 Typical Application

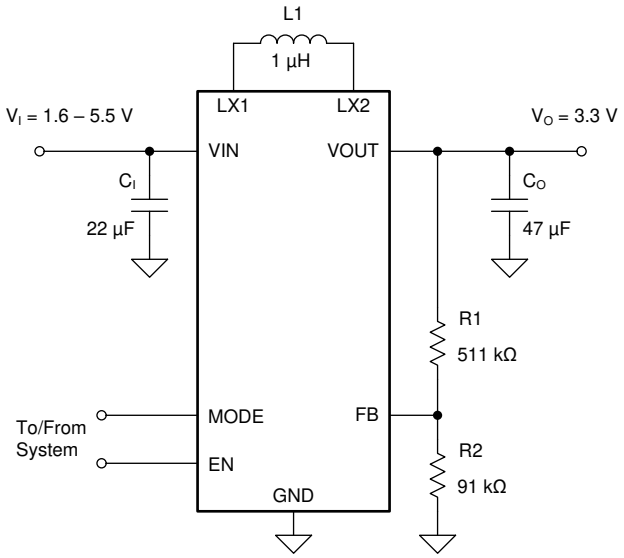


Figure 7-1. 3.3V_{OUT} Typical Application

7.2.1 Design Requirements

Table 7-1 lists the design parameters.

Table 7-1. Design Parameters

PARAMETERS	VALUES
Input voltage	2.7V to 4.3V
Output voltage	3.3V
Output current	1.5A

7.2.2 Detailed Design Procedure

The first step is the selection of the output filter components. To simplify this process, Section 5.3 outlines minimum and maximum values for inductance and capacitance. Tolerance and derating must be taken into account when selecting nominal inductance and capacitance.

7.2.2.1 Custom Design With WEBENCH® Tools

Click here to create a custom design using the TPS631000 device with the WEBENCH® Power Designer.

1. Start by entering the input voltage (V_{IN}), output voltage (V_{OUT}), and output current (I_{OUT}) requirements.
2. Optimize the design for key parameters such as efficiency, footprint, and cost using the optimizer dial.

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3. Compare the generated design with other possible solutions from Texas Instruments.

The WEBENCH Power Designer gives a customized schematic along with a list of materials with real-time pricing and component availability.

In most cases, these actions are available:

- Run electrical simulations to see important waveforms and circuit performance
- Run thermal simulations to understand board thermal performance
- Export customized schematic and layout into popular CAD formats
- Print PDF reports for the design, and share the design with colleagues

Get more information about WEBENCH tools at www.ti.com/WEBENCH.

7.2.2.2 Inductor Selection

The inductor selection is affected by several parameters such as the following:

- Inductor ripple current
- Output voltage ripple
- Transition point into power save mode
- Efficiency

See Table 7-2 for typical inductors.

For high efficiencies, the inductor must have a low DC resistance to minimize conduction losses. Especially at high-switching frequencies, the core material has a high impact on efficiency. When using small chip inductors, the efficiency is reduced mainly due to higher inductor core losses. This action must be considered when selecting the appropriate inductor. The inductor value determines the inductor ripple current. The larger the inductor value, the smaller the inductor ripple current and the lower the conduction losses of the converter. Conversely, larger inductor values cause a slower load transient response. To avoid saturation of the inductor, use Equation 3 to calculate the peak current for the inductor in steady state operation. Only the equation that defines the switch current in boost mode is shown because this provides the highest value of current and represents the critical current value for selecting the right inductor.

$$\text{Duty Cycle Boost} \qquad D = \frac{V_{OUT} - V_{IN}}{V_{OUT}} \tag{2}$$

$$I_{PEAK} = \frac{Iout}{\eta \times (1 - D)} + \frac{Vin \times D}{2 \times f \times L} \tag{3}$$

where:

- D = duty cycle in boost mode
- *f* = converter switching frequency (typical 2.2MHz)
- L = inductor value
- η = estimated converter efficiency (use the number from the efficiency curves or 0.9 as an assumption)

Note

The calculation must be done for the minimum input voltage in boost mode.

Calculating the maximum inductor current using the actual operating conditions gives the minimum saturation current of the inductor needed. TI recommends to select an inductor with a saturation current 20% higher than the value calculated using Equation 3. Table 7-2 lists possible inductors.

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Table 7-2. List of Recommended Inductors

INDUCTOR VALUE [µH]	SATURATION CURRENT [A]	DCR [mΩ]	PART NUMBER	MANUFACTURER(1)	SIZE (L × W × H mm)
1	4.3	42	DFE252012P-1R0M=P2	MuRata	2.5 × 2.0 × 1.2
1	4.2	43	HTEK20161T-1R0MSR	Cyntec	2.0 × 1.6 × 1.0
1	2.2	75	MAKK2016T1R0M (2)	Taiyo Yuden	2.0 × 1.6 × 1.0
1	2.0	144	DFE18SAN1R0ME0 (2)	Murata	1.6 × 0.8 × 0.8

(1) See the *Third-Party Products Disclaimer*.
(2) This inductor does not support full output current range.

7.2.2.3 Output Capacitor Selection

For the output capacitor, use small ceramic capacitors placed as close as possible to the VOUT and PGND pins of the IC. The recommended nominal output capacitor value is a single 47µF. If, for any reason, the application requires the use of large capacitors that cannot be placed close to the IC, use a smaller ceramic capacitor in parallel to the large capacitor. The small capacitor must be placed as close as possible to the VOUT and PGND pins of the IC.

Make sure that the effective capacitance is given according to the recommended value in Section 5.3. In general, consider DC bias effects resulting in less effective capacitance. The choice of the output capacitance is mainly a tradeoff between size and transient behavior as higher capacitance reduces transient response over/undershoot and increases transient response time. Table 7-3 lists possible output capacitors.

There is no upper limit for the output capacitance value.

Table 7-3. List of Recommended Capacitors

CAPACITOR VALUE [µF]	VOLTAGE RATING [V]	ESR [mΩ]	PART NUMBER	MANUFACTURER(1)	SIZE (METRIC)
47	6.3	10	GRM219R60J476ME44	Murata	0805 (2012)
47	10	40	CL10A476MQ8QRN	Semco	0603 (1608)

(1) See the *Third-Party Products Disclaimer*.

7.2.2.4 Input Capacitor Selection

TI recommends a 22µF input capacitor to improve line transient behavior of the regulator and EMI behavior of the total power supply circuit. An X5R or X7R ceramic capacitor placed as close as possible to the VIN and PGND pins of the IC is recommended. This capacitance can be increased without limit. If the input supply is located more than a few inches from the TPS631000, additional bulk capacitance can be required in addition to the ceramic bypass capacitors. An electrolytic or tantalum capacitor with a value of 47µF is a typical choice.

Table 7-4. List of Recommended Capacitors

CAPACITOR VALUE [µF]	VOLTAGE RATING [V]	ESR [mΩ]	PART NUMBER	MANUFACTURER(1)	SIZE (METRIC)
22	6.3	43	GRM187R61A226ME15	Murata	0603 (1608)
10	10	40	GRM188R61A106ME69	Murata	0603 (1608)

(1) See the *Third-Party Products Disclaimer*.

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7.2.2.5 Setting the Output Voltage

The output voltage is set by an external resistor divider. The resistor divider must be connected between VOUT, FB, and GND. The feedback voltage is 500mV nominal.

The low-side resistor R2 (between FB and GND) must not exceed 100kΩ. Equation 4 calculates the high-side resistor (between FB and VOUT) R1.

$$R1 = R2 \times \left(\frac{V_{OUT}}{V_{FB}} - 1 \right) \quad (4)$$

where

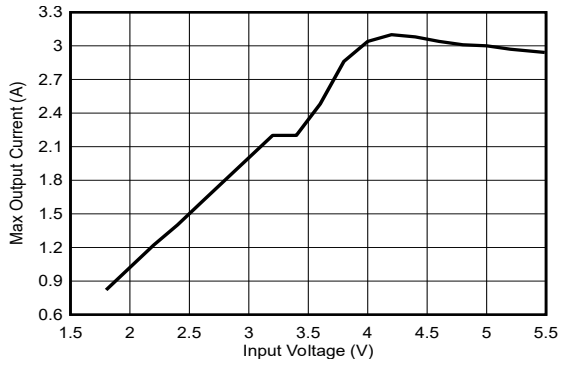
- V_{FB} = 500mV

Table 7-5. Resistor Selection For Typical Output Voltages

V_{OUT}	R1	R2
2.5V	365	91
3.3V	511	91
3.6V	562	91
5V	806	91

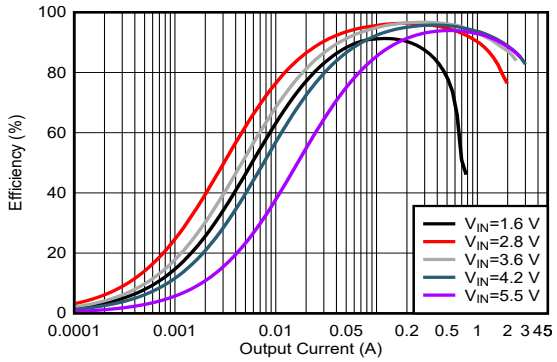
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7.2.3 Application Curves



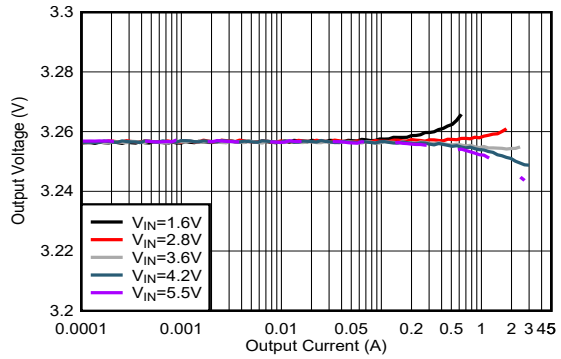
V_{OUT} = 3.3V

Figure 7-2. Typical Output Current Capability vs Input Voltage



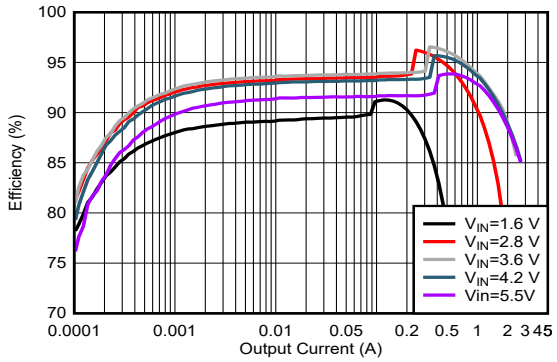
V_{OUT} = 3.3V MODE = High

Figure 7-3. Efficiency vs Output Current (FPWM)



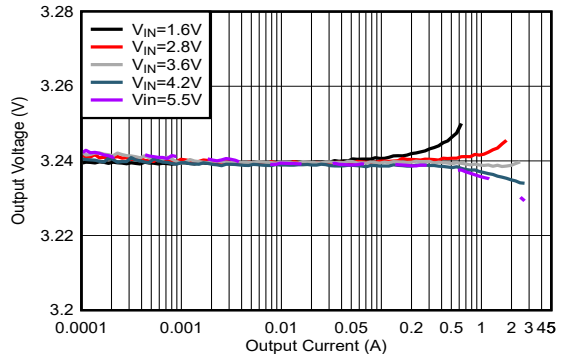
V_{OUT} = 3.3V MODE = High

Figure 7-4. Load Regulation (FPWM)



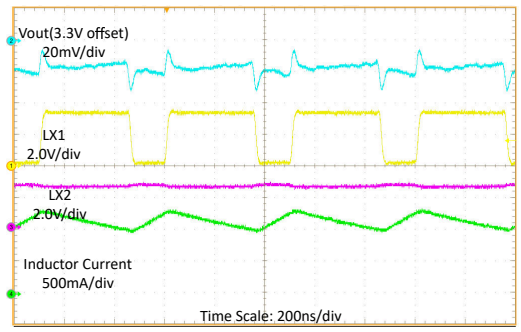
V_{OUT} = 3.3V MODE = Low

Figure 7-5. Efficiency vs Input Voltage (PFM)



V_O = 3.3V MODE = High

Figure 7-6. Load Regulation (PFM)

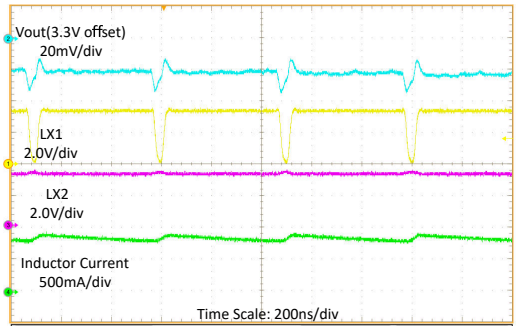


V_{IN} = 2.7V, V_{OUT} = 3.3V I_{OUT} = 1A, MODE = Low

Figure 7-7. Switching Waveforms, Boost Operation with 1A Load

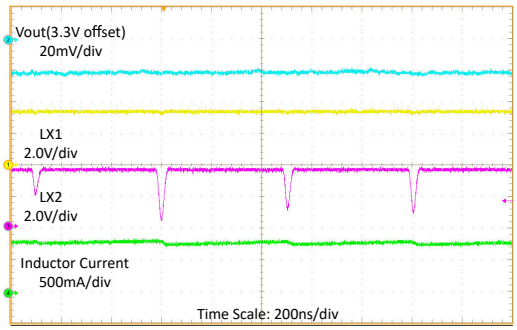
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7.2.3 Application Curves (continued)



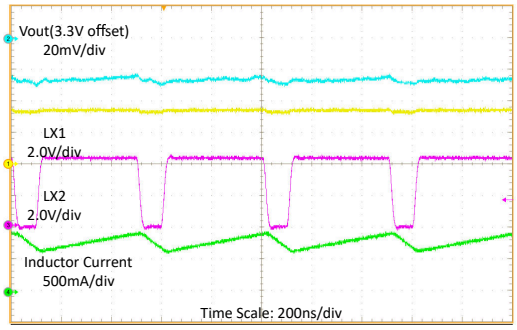
V_{IN} = 3.3V, V_{OUT} = 3.3V I_{OUT} = 1A, MODE = Low

Figure 7-8. Switching Waveforms with 1A Load



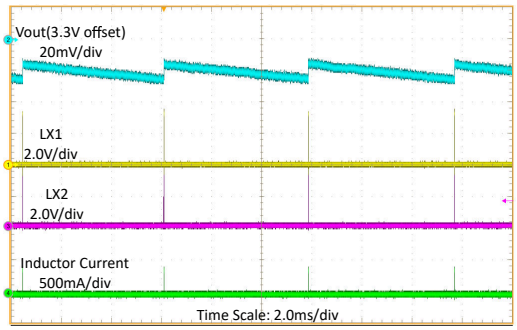
V_{IN} = 3.6V, V_{OUT} = 3.3V I_{OUT} = 1A, MODE = Low

Figure 7-9. Switching Waveforms with 1A Load



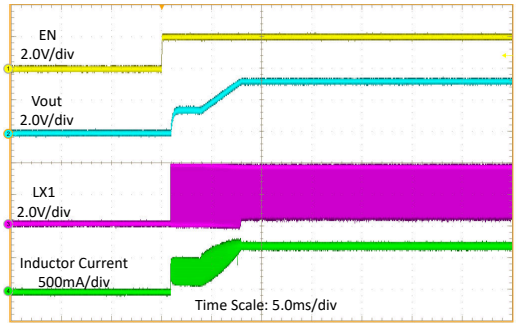
V_{IN} = 4.3V, V_{OUT} = 3.3V I_{OUT} = 1A, MODE = Low

Figure 7-10. Switching Waveforms, Buck Operation with 1A Load



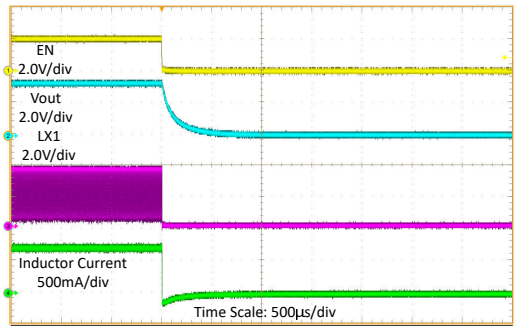
V_{IN} = 3.6V, V_{OUT} = 3.3V I_{OUT} = 1mA, MODE = Low

Figure 7-11. Switching Waveforms at 1mA Load



V_{IN} = 3.6V, V_{OUT} = 3.3V R_{load} = 4Ω, MODE = Low

Figure 7-12. Start-Up by EN

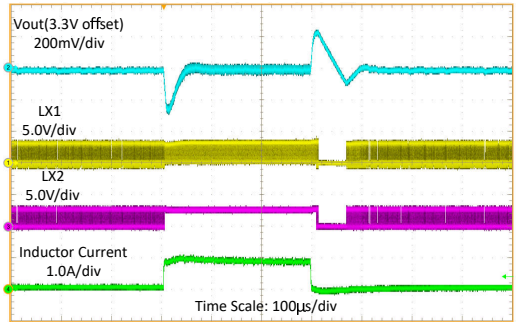


V_{IN} = 3.6V, V_{OUT} = 3.3V R_{load} = 4Ω, MODE = Low

Figure 7-13. Shutdown by EN

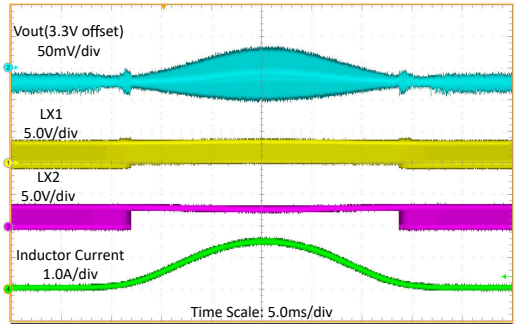
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7.2.3 Application Curves (continued)



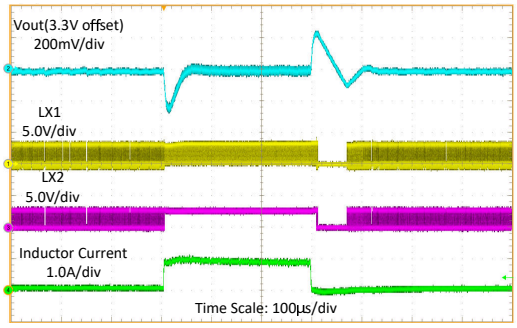
V_{IN} = 2.7V, V_{OUT} = 3.3V I_{OUT} = 100mA to 1A with 20µs slew rate

Figure 7-14. Load Transient at 2.7V Input Voltage



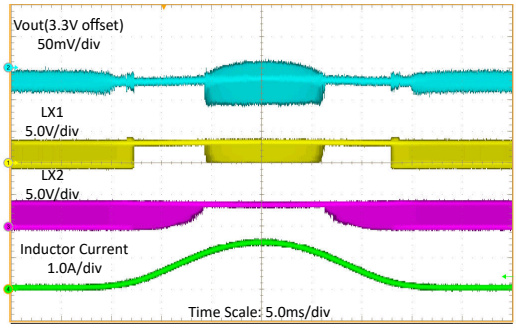
V_{IN} = 2.7V, V_{OUT} = 3.3V I_{OUT} = 100mA to 1A sweep

Figure 7-15. Load Sweep at 2.7V Input Voltage



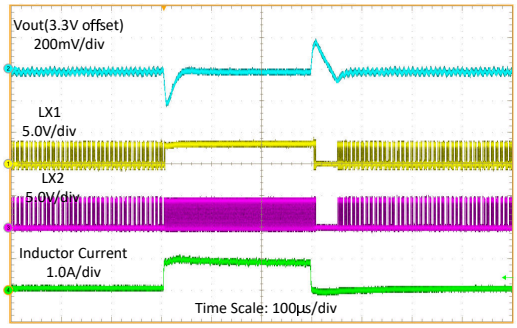
V_{IN} = 3.6V, V_{OUT} = 3.3V I_{OUT} = 100mA to 1A with 20µs slew rate

Figure 7-16. Load Transient at 3.6V Input Voltage



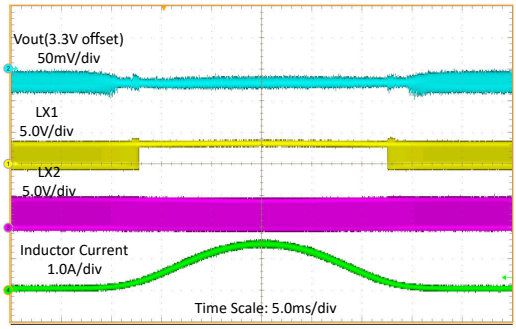
V_{IN} = 3.6V, V_{OUT} = 3.3V I_{OUT} = 100mA to 1A sweep

Figure 7-17. Load Sweep at 3.6V Input Voltage



V_{IN} = 4.3V, V_{OUT} = 3.3V I_{OUT} = 100mA to 1A with 20µs slew rate

Figure 7-18. Load Transient at 4.3V Input Voltage

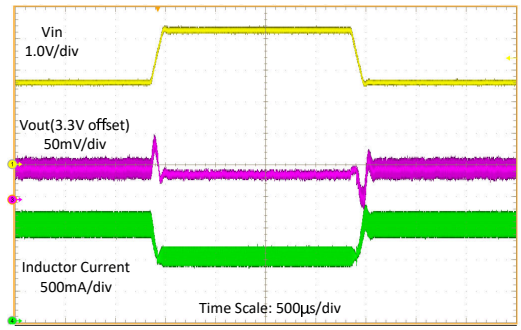


V_{IN} = 4.3V, V_{OUT} = 3.3V I_{OUT} = 100mA to 1A sweep

Figure 7-19. Load Sweep at 4.3V Input Voltage

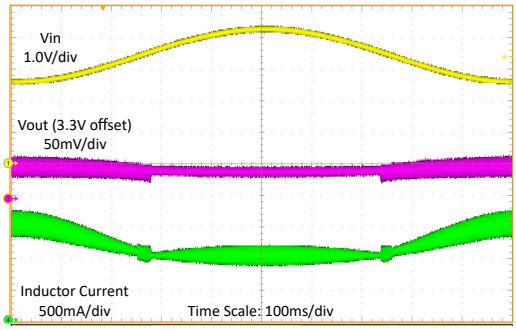
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7.2.3 Application Curves (continued)



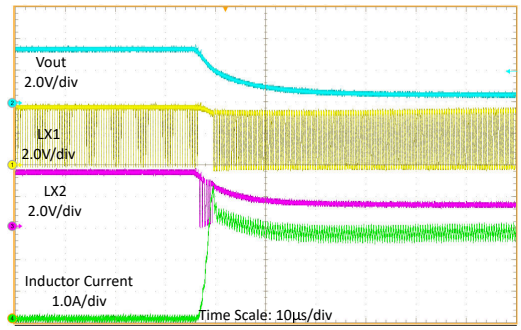
V_{IN} = 2.7V to 4.3V with 20µs slew rate, V_{OUT} = 3.3V I_{OUT} = 1A

Figure 7-20. Line Transient at 1A Load Current



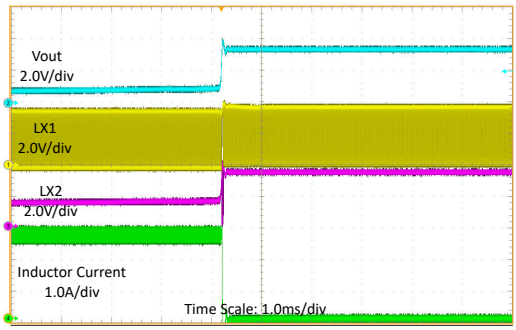
V_{IN} = 2.7V to 4.3V sweep, V_{OUT} = 3.3V I_{OUT} = 1A

Figure 7-21. Line Sweep at 1A Load Current



V_{IN} = 3.6V, V_{OUT} = 3.3V I_{OUT} = 1A, FPWM

Figure 7-22. Output Short Protection (Entry)



V_{IN} = 3.6V, V_{OUT} = 3.3V I_{OUT} = 1A, FPWM

Figure 7-23. Output Short Protection (Recover)

Table 7-6. Components for Application Characteristic Curves for V_{OUT} = 3.3V

REFERENCE	DESCRIPTION(2)	PART NUMBER	MANUFACTURER(1)
U1	High Power Density 1.5A Buck-Boost Converter	TPS631000	Texas Instruments
L1	1.0µH, 2.5mm × 2.0mm, 4.3A, 42mΩ	DFE252012P-1R0M=P2	MuRata
C1	22µF, 0603, Ceramic Capacitor, ±20%, 6.3V	GRM187R61A226ME15	Murata
C2	47µF, 0805, Ceramic Capacitor, ±20%, 6.3V	GRM219R60J476ME44	Murata
R1	511kΩ, 0603 Resistor, 1%, 100 mW	Standard	Standard
R2	91kΩ, 0603 Resistor, 1%, 100 mW	Standard	Standard

(1) See the *Third-Party Products Disclaimer*.
(2) For other output voltages, refer to Table 7-5 for resistor values.

7.3 Power Supply Recommendations

The TPS631000 device has no special requirements for the input power supply. The input power supply output current must be rated according to the supply voltage, output voltage, and output current of the TPS631000.

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7.4 Layout

7.4.1 Layout Guidelines

The PCB layout is an important step to maintain the high performance of the TPS631000.

- Place input and output capacitors as close as possible to the IC. Traces must be kept short. Route wide and direct traces to the input and output capacitors results in low trace resistance and low parasitic inductance.
- The sense trace connected to FB is signal trace. Keep these traces away from LX1 and LX2 nodes.

7.4.2 Layout Example

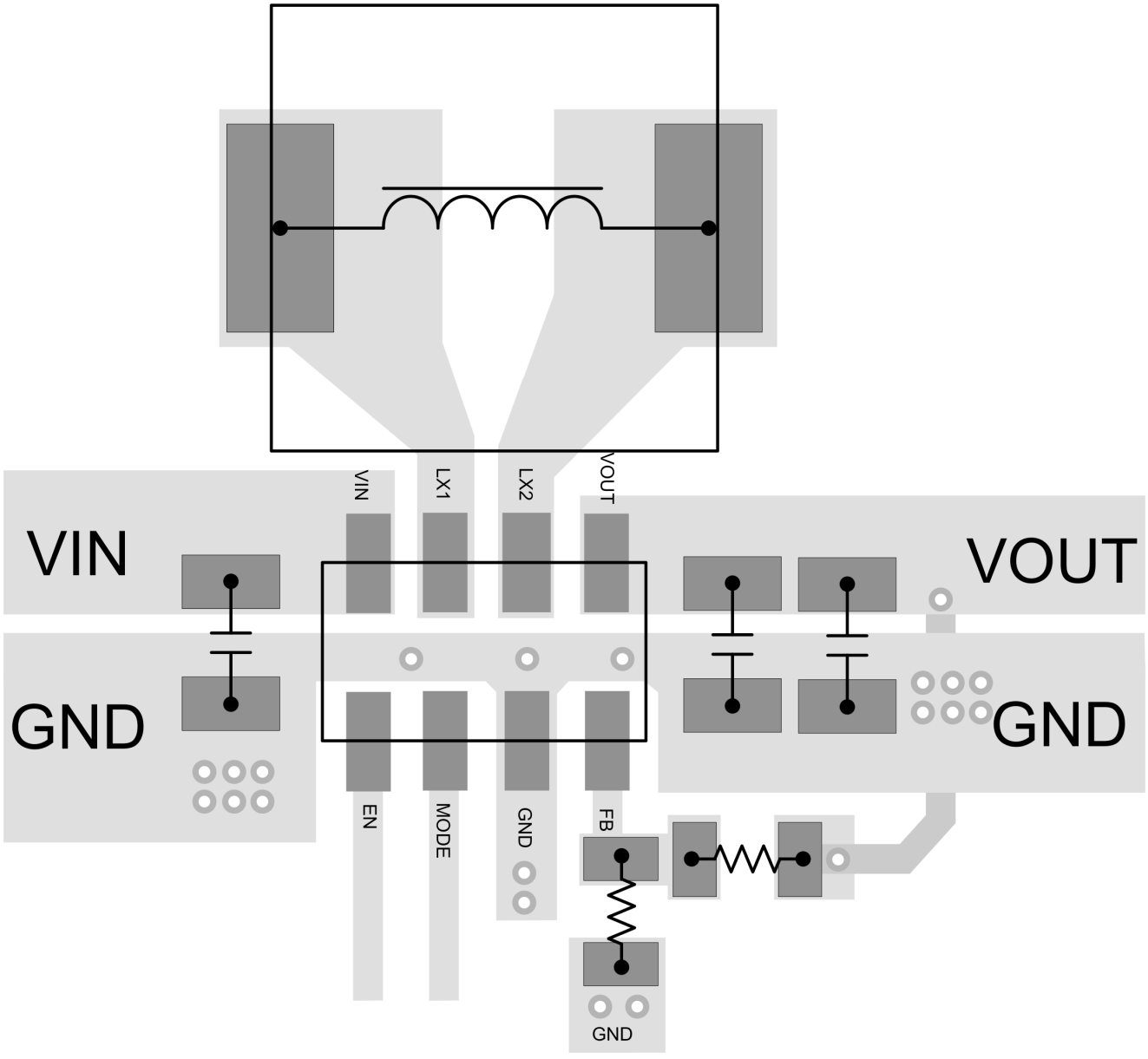


Figure 7-24. Layout Example

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8 Device and Documentation Support

8.1 Device Support

8.1.1 Third-Party Products Disclaimer

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8.1.2 Development Support

8.1.2.1 Custom Design With WEBENCH® Tools

Click here to create a custom design using the TPS631000 device with the WEBENCH® Power Designer.

1. Start by entering the input voltage (V_{IN}), output voltage (V_{OUT}), and output current (I_{OUT}) requirements.
2. Optimize the design for key parameters such as efficiency, footprint, and cost using the optimizer dial.
3. Compare the generated design with other possible solutions from Texas Instruments.

The WEBENCH Power Designer gives a customized schematic along with a list of materials with real-time pricing and component availability.

In most cases, these actions are available:

- Run electrical simulations to see important waveforms and circuit performance
- Run thermal simulations to understand board thermal performance
- Export customized schematic and layout into popular CAD formats
- Print PDF reports for the design, and share the design with colleagues

Get more information about WEBENCH tools at www.ti.com/WEBENCH.

8.2 Receiving Notification of Documentation Updates

To receive notification of documentation updates, navigate to the device product folder on ti.com. Click on *Notifications* to register and receive a weekly digest of any product information that has changed. For change details, review the revision history included in any revised document.

8.3 Support Resources

TI E2E™ support forums are an engineer's go-to source for fast, verified answers and design help — straight from the experts. Search existing answers or ask your own question to get the quick design help you need.

Linked content is provided "AS IS" by the respective contributors. They do not constitute TI specifications and do not necessarily reflect TI's views; see TI's Terms of Use.

8.4 Trademarks

TI E2E™ is a trademark of Texas Instruments.
WEBENCH® is a registered trademark of Texas Instruments.
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8.5 Electrostatic Discharge Caution



This integrated circuit can be damaged by ESD. Texas Instruments recommends that all integrated circuits be handled with appropriate precautions. Failure to observe proper handling and installation procedures can cause damage.

ESD damage can range from subtle performance degradation to complete device failure. Precision integrated circuits may be more susceptible to damage because very small parametric changes could cause the device not to meet its published specifications.

8.6 Glossary

TI Glossary	This glossary lists and explains terms, acronyms, and definitions.

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9 Revision History

NOTE: Page numbers for previous revisions may differ from page numbers in the current version.

Changes from Revision B (June 2022) to Revision C (August 2026)	Page
• Changed the min value of "Input voltage for less than 10ns (LX1, LX2)" from –0.3 to –2 in the *Absolute Maximum Ratings table*	4
• Changed the CDM standard from "JEDEC specification JESD22C101" to "ANSI/ESDA/JEDEC JS-002"	4

Changes from Revision A (January 2022) to Revision B (June 2022)	Page
• Removed reference to PSM	7
• Updated Table 7-2	11
• Updated Table 7-6	14

Changes from Revision * (October 2021) to Revision A (January 2022)	Page
• Changed document status from Advance Information to Production Data	1

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10 Mechanical, Packaging, and Orderable Information

The following pages include mechanical, packaging, and orderable information. This information is the most current data available for the designated devices. This data is subject to change without notice and revision of this document. For browser-based versions of this datasheet, see also the left-hand navigation.

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PACKAGING INFORMATION

Orderable part number	Status (1)	Material type (2)	Package | Pins	Package qty | Carrier	RoHS (3)	Lead finish/ Ball material (4)	MSL rating/ Peak reflow (5)	Op temp (°C)	Part marking (6)
TPS631000DRLR	Active	Production	SOT-5X3 (DRL) | 8	4000 | LARGE T&R	Yes	Call TI | Sn	Level-1-260C-UNLIM	-40 to 125	2N4W
TPS631000DRLR.A	Active	Production	SOT-5X3 (DRL) | 8	4000 | LARGE T&R	Yes	SN	Level-1-260C-UNLIM	-40 to 125	2N4W

(1) **Status:** For more details on status, see our product life cycle.

(2) **Material type:** When designated, preproduction parts are prototypes/experimental devices, and are not yet approved or released for full production. Testing and final process, including without limitation quality assurance, reliability performance testing, and/or process qualification, may not yet be complete, and this item is subject to further changes or possible discontinuation. If available for ordering, purchases will be subject to an additional waiver at checkout, and are intended for early internal evaluation purposes only. These items are sold without warranties of any kind.

(3) **RoHS values:** Yes, No, RoHS Exempt. See the TI RoHS Statement for additional information and value definition.

(4) **Lead finish/Ball material:** Parts may have multiple material finish options. Finish options are separated by a vertical ruled line. Lead finish/Ball material values may wrap to two lines if the finish value exceeds the maximum column width.

(5) **MSL rating/Peak reflow:** The moisture sensitivity level ratings and peak solder (reflow) temperatures. In the event that a part has multiple moisture sensitivity ratings, only the lowest level per JEDEC standards is shown. Refer to the shipping label for the actual reflow temperature that will be used to mount the part to the printed circuit board.

(6) **Part marking:** There may be an additional marking, which relates to the logo, the lot trace code information, or the environmental category of the part.

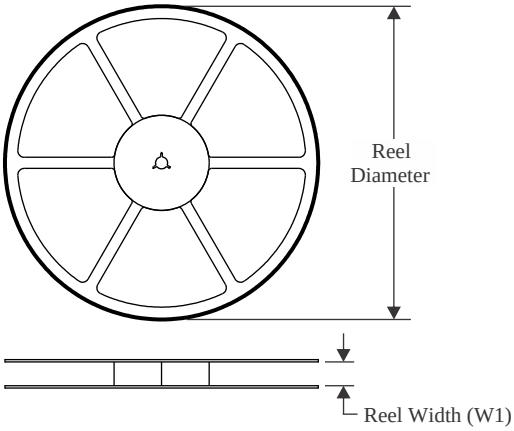
Multiple part markings will be inside parentheses. Only one part marking contained in parentheses and separated by a "~" will appear on a part. If a line is indented then it is a continuation of the previous line and the two combined represent the entire part marking for that device.

Important Information and Disclaimer: The information provided on this page represents TI's knowledge and belief as of the date that it is provided. TI bases its knowledge and belief on information provided by third parties, and makes no representation or warranty as to the accuracy of such information. Efforts are underway to better integrate information from third parties. TI has taken and continues to take reasonable steps to provide representative and accurate information but may not have conducted destructive testing or chemical analysis on incoming materials and chemicals. TI and TI suppliers consider certain information to be proprietary, and thus CAS numbers and other limited information may not be available for release.

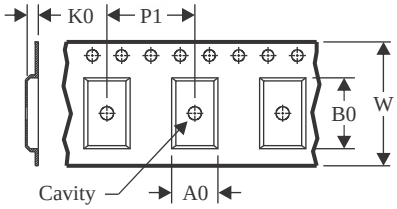
In no event shall TI's liability arising out of such information exceed the total purchase price of the TI part(s) at issue in this document sold by TI to Customer on an annual basis.

TAPE AND REEL INFORMATION

REEL DIMENSIONS

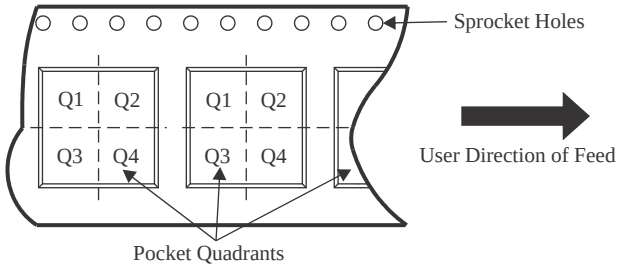


TAPE DIMENSIONS



A0	Dimension designed to accommodate the component width
B0	Dimension designed to accommodate the component length
K0	Dimension designed to accommodate the component thickness
W	Overall width of the carrier tape
P1	Pitch between successive cavity centers

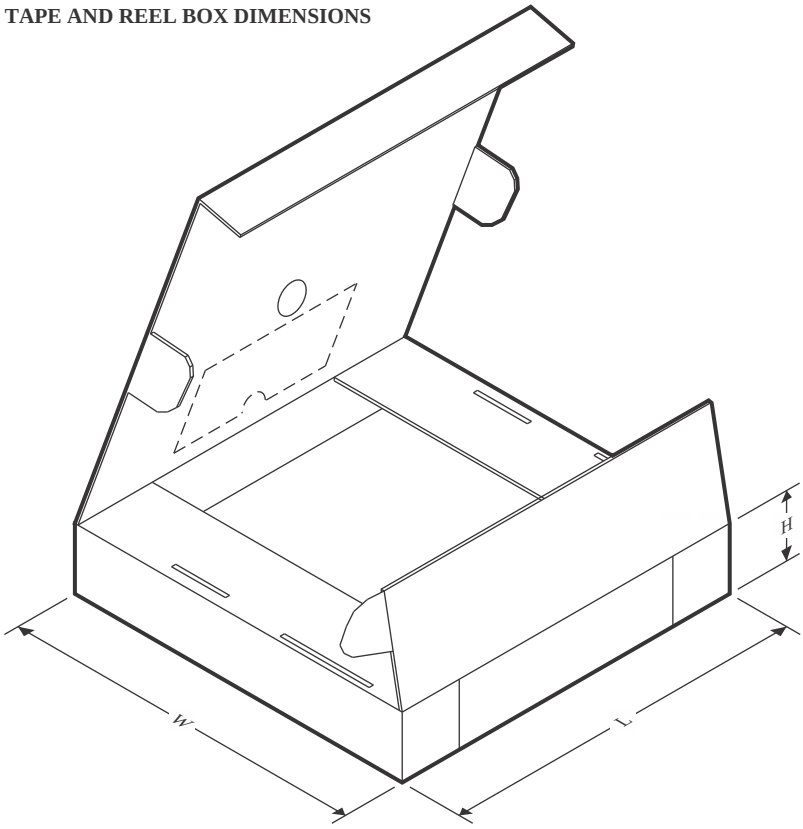
QUADRANT ASSIGNMENTS FOR PIN 1 ORIENTATION IN TAPE



*All dimensions are nominal

Device	Package Type	Package Drawing	Pins	SPQ	Reel Diameter (mm)	Reel Width W1 (mm)	A0 (mm)	B0 (mm)	K0 (mm)	P1 (mm)	W (mm)	Pin1 Quadrant
TPS631000DRLR	SOT-5X3	DRL	8	4000	180.0	8.4	2.75	1.9	0.8	4.0	8.0	Q3

TAPE AND REEL BOX DIMENSIONS



*All dimensions are nominal

Device	Package Type	Package Drawing	Pins	SPQ	Length (mm)	Width (mm)	Height (mm)
TPS631000DRLR	SOT-5X3	DRL	8	4000	210.0	185.0	35.0

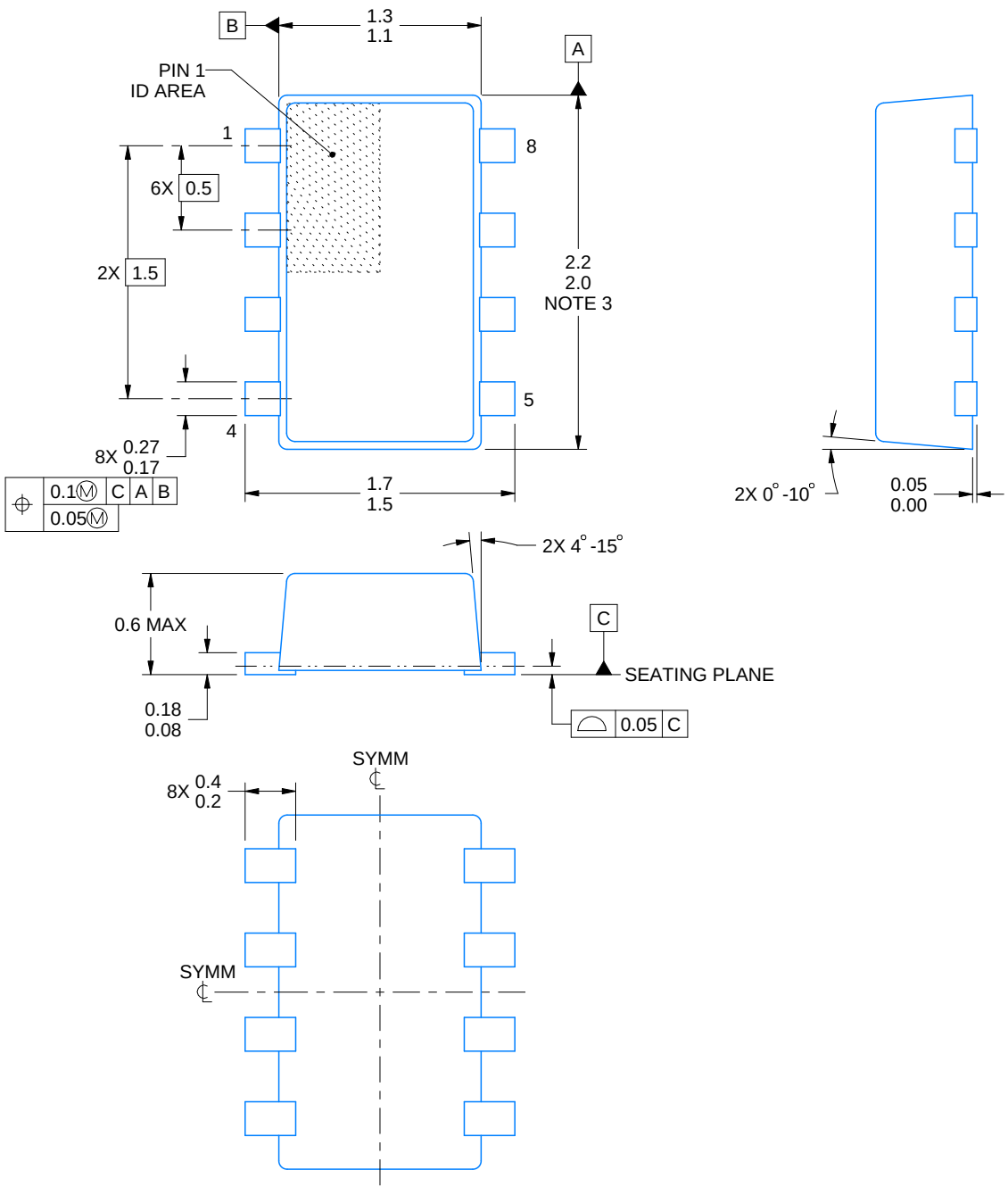


PACKAGE OUTLINE

DRL0008A

SOT-5X3 - 0.6 mm max height

PLASTIC SMALL OUTLINE



4224486/G 11/2024

NOTES:

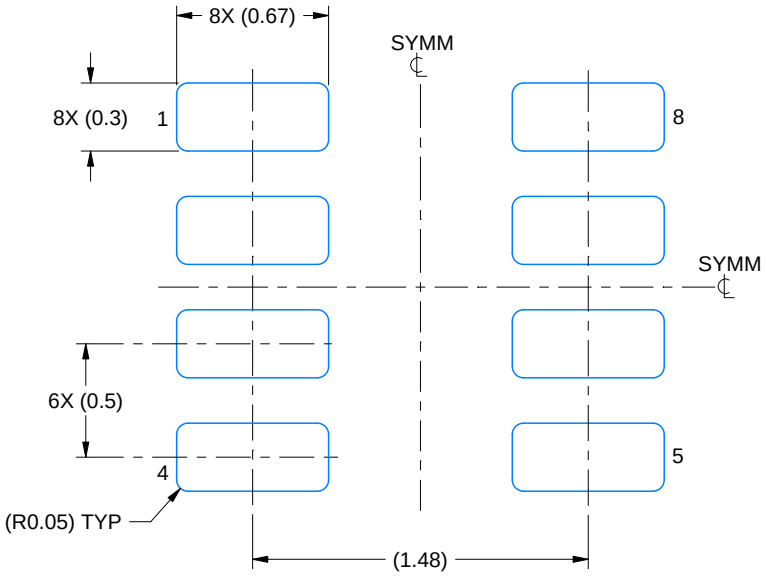
1. All linear dimensions are in millimeters. Any dimensions in parenthesis are for reference only. Dimensioning and tolerancing per ASME Y14.5M.
2. This drawing is subject to change without notice.
3. This dimension does not include mold flash, protrusions, or gate burrs. Mold flash, interlead flash, protrusions, or gate burrs shall not exceed 0.15 mm per side.
4. Reference JEDEC Registration MO-293, Variation UDAD

EXAMPLE BOARD LAYOUT

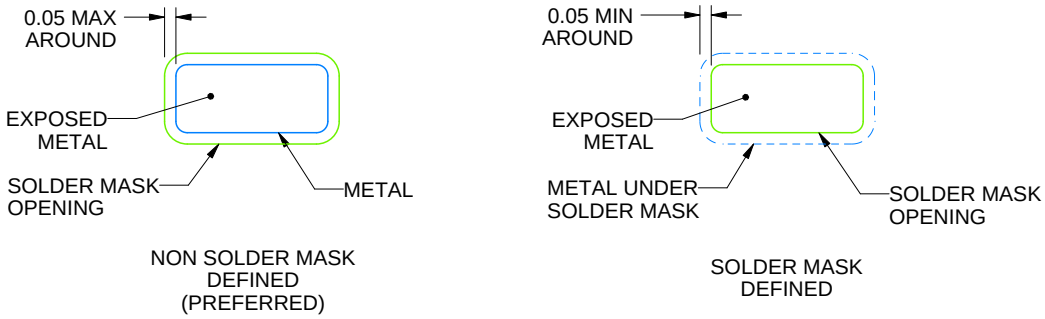
DRL0008A

SOT-5X3 - 0.6 mm max height

PLASTIC SMALL OUTLINE



LAND PATTERN EXAMPLE
EXPOSED METAL SHOWN
SCALE:30X



SOLDERMASK DETAILS

4224486/G 11/2024

NOTES: (continued)

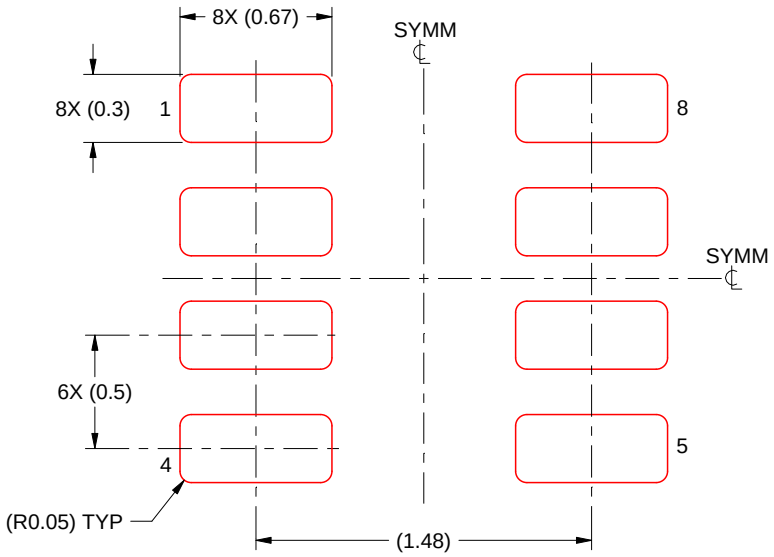
5. Publication IPC-7351 may have alternate designs.
6. Solder mask tolerances between and around signal pads can vary based on board fabrication site.
7. Land pattern design aligns to IPC-610, Bottom Termination Component (BTC) solder joint inspection criteria.

EXAMPLE STENCIL DESIGN

DRL0008A

SOT-5X3 - 0.6 mm max height

PLASTIC SMALL OUTLINE



SOLDER PASTE EXAMPLE
BASED ON 0.1 mm THICK STENCIL
SCALE:30X

4224486/G 11/2024

NOTES: (continued)

8. Laser cutting apertures with trapezoidal walls and rounded corners may offer better paste release. IPC-7525 may have alternate design recommendations.
9. Board assembly site may have different recommendations for stencil design.

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Last updated 10/2025